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Q:

Explain the static and dynamic approach of Software Testing?

In Software Testing, both static and dynamic approaches are used to identify defects and ensure software quality. These approaches differ in how they are applied and the stage at which they are used in the software development lifecycle (SDLC).

1. Static Testing (Pre-Execution)

Definition: Static testing is a **preventive** approach that involves reviewing and analyzing software artifacts (e.g code, design, documents, requirements) without executing the program.

Key Aspects:-

→ Conducted **early** in the SDLC to detect defects before execution.

→ Includes techniques like **reviews, walk throughs, and static code analysis.**

→ Helps in finding issues like **syntax errors, missing requirements, coding standards violations, and documentation defects.**

Types of Static Testing:

1. **Reviews** – Examination of documents (e.g requirement reviews, code reviews) by team members.
2. **Walk throughs** – Informal step by step presentations of a document or code by the author to gather feedback.
3. **Inspection** – A more formal and detailed review of documents or code by experts.
4. **Static Code Analysis** – Using tools (e.g SonarQube, ESlint, Checkstyle) to analyze source code for vulnerabilities, unused variables and syntax [illegible]

Advantages of Static Testing:

* Early detection of defects reduces cost and effort.
* Helps ensure coding standards and best practices.
* Improves documentation and design quality.

Dynamic Testing (Execution-Based)

Definition: Dynamic testing involves **executing the software** to validate its functionality, performance, and behaviour.

Key Aspects:

- Performed during or after software Development.
- Includes different types of testing like **function performance, security and usability testing.**
- Helps identify **runtime errors, logical issues and system behavior under different condi**

Advantage: 1. Identify real world issues

2. end to end system functionality validate

Date

Types of Dynamic Testing:

1. **Functional Testing**: Validates if the system behaves as expected (e.g. unit testing, integration testing, system testing, acceptance testing).

2. **Non functional Testing**: Evaluates performance, security, usability and reliability (e.g. load testing, Stress testing, security testing).

3. **White Box Testing**: Examines internal code structure and logic

4. **Black Box Testing**: Focuses on input/output without knowledge of internal code.

5. **Grey Box Testing**: Combines white box and black box approaches

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